

OPERATING SYSTEMS (CS304)

10 Marks Question Bank

UNIT 1 – Processes, Threads & System Calls

1. Explain the Process Life Cycle with a neat diagram.
2. Discuss Process Control Block (PCB) and its importance.
3. Differentiate Processes and Threads with examples.
4. Explain System Calls and their various categories.

UNIT 2 – CPU Scheduling

1. Explain FCFS, SJF, and Round Robin Scheduling Algorithms with examples.
2. Discuss Scheduling Criteria and performance measures.
3. Solve a CPU Scheduling problem using Round Robin algorithm.
4. Compare Preemptive and Non-Preemptive Scheduling.

UNIT 3 – Deadlocks

1. Define Deadlock and explain the necessary conditions for its occurrence.
2. Explain Deadlock Prevention and Avoidance techniques.
3. Describe the Banker's Algorithm with an example.
4. Discuss Resource Allocation Graphs and Deadlock Detection.

UNIT 4 – Memory Management

1. Explain Paging with suitable examples.
2. Discuss Segmentation and its advantages.
3. Differentiate Paging and Segmentation.
4. Explain Virtual Memory and Thrashing.

UNIT 5 – File Systems & Disk Scheduling

1. Explain various File Allocation Methods.
2. Discuss Directory Structures in Operating Systems.
3. Explain FCFS, SSTF, SCAN, and C-SCAN Disk Scheduling Algorithms.
4. Compare different Disk Scheduling Algorithms with examples.

Exam Preparation Tips

Focus on Process Management, CPU Scheduling Numerical Problems, Deadlock Handling, Memory Management Techniques, and Disk Scheduling Algorithms as these are frequently asked 10-mark university questions.